

# C++17 should refer to ISO/IEC 10646 2014 instead of 1994

ISO standards are only supposed to have normative references to the latest version of other ISO standards, yet the C++17 CD still refers to *ISO/IEC 10646-1:1993, Information technology — Universal Multiple-Octet Coded Character Set (UCS)— Part 1: Architecture and Basic Multilingual Plane*.

There have been three revisions and numerous amendments of 10646 since 1994. The changes that impact the C++17 CD include:

- The name has changed to *Information technology — Universal Coded Character Set (UCS)*.
- UTF-8, UTF-16, and UTF-32 are now defined. They were not even a part of 10646:1994 before amendments, so the C++ standard has been using the terms without a normative definition.
- UCS-2 has been deprecated, and has been replaced by UTF-16. This is a normative change for the C++ standard because UCS-2 and UTF-16 are not the same.
- The term UCS-4 has been changed to UTF-32. Although 10646 says "The terms UTF-32 and UCS-4 can be used interchangeably...", the C++ standard should use the preferred term UTF-32 throughout.
- New characters have been added. Annex E needs to be updated accordingly.

See <http://standards.iso.org/ittf/PubliclyAvailableStandards/index.html> for a copy of ISO/IEC 10646:2014.

## Proposed changes

Strike the wording *high-lighted in red* and add the wording *high-lighted in green*.

### 1.2 Normative references [intro.refs]

— ISO/IEC 10646-1:1993, *Information technology — Universal Multiple-Octet Coded Character Set (UCS) — Part 1: Architecture and Basic Multilingual Plane* :2014, *Information technology — Universal Coded Character Set (UCS)*

### 22.5 Standard code conversion facets [locale.stdcvt]

For the facet `codecvt_utf8`:

— The facet shall convert between UTF-8 multibyte sequences and `UCS2` `UTF-16` or `UCS4` `UTF-32` (depending on the size of `Elem`) within the program.

...

For the facet `codecvt_utf16`:

— The facet shall convert between UTF-16 multibyte sequences and `UCS2` `UTF-16` or `UCS4` `UTF-32` (depending on the size of `Elem`) within the program.

### E.1 Ranges of characters allowed [charname.allowed]

10000-1FFFD, 20000-2FFFD, 30000-3FFFD, 40000-4FFFD, 50000-5FFFD,  
60000-6FFFD, 70000-7FFFD, 80000-8FFFD, 90000-9FFFD, A0000-AFFFD,  
B0000-BFFFD, C0000-CFFFD, D0000-DFFFD, E0000-EFFFD, F0000-FFFFD,  
100000-10FFFFD

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